

Descriptive Epidemiology of Musculoskeletal Injuries Presenting to an Embedded Holistic Health and Fitness (H2F) Team in a United States Army Stryker Brigade, October 2021-September 2022

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ABSTRACT

Introduction:

In 2020, the U.S. Army began implementation of the Holistic Health and Fitness (H2F) system to reduce musculoskeletal injury (MSKI) rates and expedite rehabilitation after injury. The purpose of this study was to describe the rates and characteristics of incident MSKIs evaluated by H2F physical therapists (PTs) and athletic trainers (ATs) in a U.S. Army Stryker brigade combat team from October 2021 to September 2022.

Materials and Methods:

In this retrospective observational study, we used negative binomial models and the H2F injury tracker and Army administrative data to estimate MSKI incidence rates for the overall brigade and within strata of battalion, age, sex, rank, race, and ethnicity. We repeated these estimates when standardized to the overall brigade population by battalion, age, sex, and rank. We characterized distributions of time from injury to evaluation, limited duty days (i.e., time-loss), and activity at time of injury. This research was determined exempt by academic and military Institutional Review Boards.

Results:

The embedded H2F PTs and ATs evaluated 2,140 incident MSKIs in the brigade (4,780 person-years) for an overall rate of 39.1 MSKIs per 1,000 person-months. Battalion rates ranged from 31.8 (field artillery) to 50.0 (infantry) and did not change substantially after standardization. Embedded H2F PTs and ATs evaluated 63% of MSKIs within 1 week of onset, managed 59% without assigning time-loss, and found 51% of MSKIs occurred during unit physical fitness training.

Conclusions:

These findings are consistent with the a priori expectation that embedded providers and the H2F system expedited access to MSKI evaluation and treatment. A strength of this study was our ability to estimate MSKI incidence rates and characterize MSKI evaluated in the H2F environment using a novel injury tracker coupled with administrative data. A limitation is that because of data deidentification, we could not measure cumulative time-loss. This descriptive observational study will help inform the findings of subsequent research on the effect of H2F implementation on MSKI outcomes.

INTRODUCTION

The surveillance and prevention of musculoskeletal injury (MSKI) in the U.S. Army is a critical issue for military public health and defense readiness.¹ In 2018 alone, MSKIs accounted

for 8 million limited duty days (i.e., time-loss) and \$434 million in direct patient care costs in the active U.S. Army.² Estimates that included indirect medical and limited duty day costs for that same year ranged from 2.3 to 4.0 billion dollars.³ The most recent

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The findings and conclusions of this research are those of the authors and do not represent the official position of the U.S. Army Medical Center

of Excellence, the U.S. Army Training and Doctrine Command, 5th Security Force Assistance Brigade, 10th Special Forces Group (Airborne), the Defense Health Agency, Department of the Army, Department of Defense, the CDC, or any agency in the U.S. Government.

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doi:<https://doi.org/10.1093/milmed/usaf645>

Published by Oxford University Press on behalf of the Association of Military Surgeons of the United States 2026. This work is written by (a) US Government employee(s) and is in the public domain in the US.

Defense Centers for Public Health—Aberdeen (DCPH-A) Annual Injury Surveillance Report indicated MSKIs accounted for 81% of all incident injuries sustained in the active U.S. Army in 2022.⁴

Injury prevention professionals have advocated for militaries to embed health and human performance teams, similar to the model of embedded health/performance providers found in collegiate and professional athletics. In 2020, the U.S. Army began implementing the Holistic Health and Fitness (H2F) system, an initiative designed to improve the health and readiness of soldiers across multiple domains of health.⁵ H2F was modeled after programs previously implemented in special operations units to expedite access to MSKI evaluation and management.^{5–7} To date, few studies have examined service utilization in H2F environments, and none have reported MSKI rates.^{8–10}

This study addresses this gap in the literature by describing rates and characteristics of MSKIs evaluated by an H2F team over a 12-month period in a U.S. Army Stryker brigade combat team located in the continental U.S. We estimated MSKI rates for the brigade and stratified by battalion type and important demographics, with the goal of understanding whether observed differences in MSKI rates between groups were explained by demographic differences. To contextualize these rates, we estimated the overall proportion of MSKIs evaluated by the H2F team compared with all brigade MSKIs recorded in electronic health records (EHR). We also characterized MSKIs evaluated by H2F healthcare providers, including days from MSKI onset to evaluation, limited duty days assigned, and MSKIs associated with unit physical fitness training.

METHODS

Study Design and Participants

We conducted a retrospective observational open-cohort study of a Stryker brigade combat team. The dynamic cohort comprised all active-duty individuals assigned to any of the brigade's 7 battalions (3 infantry battalions and one battalion each of cavalry, field artillery, engineer, and support) for 1 or more months during the 12-month period from October 2021 to September 2022. Our results are reported in accordance with the STrengthening the Reporting of OBservational Studies in Epidemiology (STROBE) Statement.¹¹

DATA SOURCES

We used 3 data sources for this study. MSKI data were abstracted from an injury tracker database created by the brigade H2F team (described below). This custom data resource was supplemented by 2 others: aggregated demographic data from administrative records to quantify person-time at risk and external EHR used to estimate the proportion of brigade MSKIs evaluated by H2F.

Internal Injury Tracker

This database recorded all incident MSKIs in the brigade evaluated by H2F healthcare providers. Individuals with MSKI

presented for evaluation via direct access or after referral by battalion medics, physician assistants, unit leaders, or external medical personnel. During the study period there were 3 physical therapists (PTs) and 4–6 athletic trainers (ATs) that evaluated MSKIs for the brigade. As ATs were hired, they were each emplaced within a battalion, beginning with infantry battalions, and typically evaluated and treated MSKI from a battalion training center. The 3 PTs were responsible for one or more battalions and located within a centralized brigade training center. PTs evaluated and treated MSKIs from these battalions and coordinated care with the battalion ATs. PTs ordered imaging, prescribed medications, or wrote referrals to other specialty care (e.g., orthopedics) as needed. Generally, ATs and PTs conducted MSKI evaluation as part of morning sick call in brigade or battalion training center spaces, as that was where injured individuals went in lieu of, or upon injury during physical training. Outside of sick call, H2F healthcare providers could evaluate or schedule evaluations at any time during the workday. Although H2F also included occupational therapists (OTs), MSKI evaluations performed by OTs were not recorded on the injury tracker during the study period.

H2F staff recorded clinical characteristics for each MSKI evaluation, including: body region injured and laterality, activity during which injury occurred (unit military training [e.g., falling and spraining wrist during field exercise], unit physical training [e.g., developing anterior knee pain during unit run], work-related nontraining [e.g., straining muscle when lifting a box not specific to military training], personal time [e.g., spraining ankle when running on the weekend], and nonspecific onset), whether the patient reported their injury had caused them to “miss or modify work or training” before evaluation (yes/no), time from injury to evaluation (days), initial time-loss assigned by H2F provider (yes/no), duration of initial time-loss (days), and H2F evaluating provider type (AT, PT). The presenting soldier's DoD Identification number (DoD ID) was used to query demographic information recorded in U.S. Army systems.

We defined incident MSKI as the first recorded injury to a body region (e.g., knee) and laterality (e.g., right, left). Subsequent MSKI of the same body region and laterality (e.g., right knee) were excluded from analysis. MSKI to a different body region or laterality was included. We used a lookback period of the prior 3 months to avoid counting proximal reinjury or exacerbation as incident MSKI during the study period.¹²

Person-Time at Risk

We used deidentified demographic data from the Defense Manpower Data Center (DMDC) to quantify person-time of the total population at risk. These data were aggregated monthly across the following variables: unit identification code (UIC; 6-digit alphanumeric identifier of company), age (years), sex (female, male), race (1 of 6: American Indian or Alaska Native; Asian; Black or African American; Native Hawaiian or Pacific Islander; White; or Unknown/Not Applicable), ethnicity (Hispanic or Latino, Not Hispanic or Latino), and rank (grade; e.g., E-1, E-2).

We used a list of UICs from Human Resources Command to group person-time by battalion. We included the brigade headquarters element with the support battalion and the forward support companies from the support battalion with the battalions to which they were assigned. We categorized age and rank similar to published injury surveillance reports in the Army,¹³ with the uppermost categories collapsed because of smaller numbers in the sample.

External Electronic Health Record Data

Total incident MSKI counts from the Defense Medical Surveillance System (DMSS) were used to estimate the proportion of all brigade-level incident MSKI evaluated by the H2F team. This limited data set containing encounters for all MSKI diagnoses per the DCPH-A injury taxonomy¹⁴ in the primary position was obtained from the Armed Forces Health Surveillance Division (AFHSD) of DHA.

ETHICS APPROVAL

This research was determined exempt by the Institutional Review Board of the U.S. Army Medical Center of Excellence and the University of North Carolina at Chapel Hill. Use of these data sets for research purposes was endorsed by the unit commander, H2F team, and DHA, and the requisite data-sharing agreements were enacted. Requests for these data must be made through DHA and the U.S. Army.

STATISTICAL ANALYSIS

Descriptive statistics for MSKI characteristics were calculated (e.g., proportion, median, and interquartile range [IQR]). Because of overdispersion, we used negative binomial models to estimate unstandardized and standardized MSKI incidence rates, rate ratios, and 95% confidence intervals (CIs). We used incident MSKI as the count outcome and the natural log of person-time as the model offset. We estimated the overall brigade MSKI rate and stratum-specific rates and rate ratios for battalion, age, sex, rank, race, and ethnicity. We repeated these estimates when standardized to the overall marginal distributions of battalion, age, sex, and rank in the brigade as a whole. For example, when estimating standardized MSKI rates by battalion, we controlled for battalion differences in the distribution of age, sex, and rank, covariates associated with MSKI.^{15–18} We did not include terms for race or ethnicity when standardizing for reasons described elsewhere.¹⁹

Missing Data

Some MSKIs were missing data for the following demographic variables: age, sex, and race (7% each); ethnicity (10%); and rank (12%) (Table S1). Most of these were believed to originate from minor transcription errors by H2F staff (e.g., DoD ID recorded erroneously). We assumed this missingness primarily arose from Missing at Random (MAR) mechanisms²⁰ and used Multiple Imputation by Chained Equations (MICE) to create 20 imputed data sets.^{21,22} The rates and rate ratios presented

here are pooled results^{20,23} from the 20 imputed analyses. All analyses were conducted in SAS 9.4 (Cary, NC).

RESULTS

The study population included 4,780 person-years over the 12-month study period, consistent with the expected brigade size (4,500–5,000 personnel). Battalions were similar across most demographic variables, with some notable differences (Table 1). Engineer and support battalions had the highest percentage of female personnel, approaching 20%, while infantry and cavalry battalions had the lowest (<10%). Most infantry battalion personnel were <25 years old (53%), junior ranking (63%), and White (74%). The support battalion, which included brigade headquarters, had the highest percentage of personnel >35 years old, senior ranked, Black or African American, Asian, and Native Hawaiian or Pacific Islander.

MSKI Incidence Rates

H2F healthcare providers evaluated 2,140 incident MSKIs sustained by 1,722 individuals over 12 months (Table 2). The overall brigade MSKI incidence rate was 39.1 MSKIs/1,000 person-months (p-m) (95% CI: 35.7, 42.9). Infantry battalions had the highest rate (50.0 MSKIs/1,000 p-m; 95% CI: 42.7, 58.4), and field artillery had the lowest (31.8 MSKIs/1,000 p-m; 95% CI: 25.3, 40.0) (Figure 1). Females demonstrated a lower rate than males, though the estimate was likely less precise because of smaller numbers. For age, estimates varied substantially between the youngest (<25) and older categories (25–34, ≥35). Senior noncommissioned officers (NCOs) and officers (commissioned and warrant) had a rate approximately 40% less than junior enlisted and NCO ranks (Table S2). For known race categories, Asian had the lowest MSKI incidence rates, followed by American Indian or Alaska Native; Black or African American and White had similar rates; and Native Hawaiian or Pacific Islander had the highest rates. Native Hawaiian or Pacific Islander and American Indian or Alaska Native had the most imprecise estimates, likely because of comprising the smallest percentages of the total population. For ethnicity, MSKI rates were higher in the Hispanic or Latino category compared with Not Hispanic or Latino.

There was little relative change to most estimates when standardizing to the population distributions of battalion type, age, sex, and rank. For strata where estimates increased, this was possibly from adopting a higher risk profile based on brigade composition (mostly infantry, male, <25, junior enlisted) (Figure 1).

The EHR data set included 4,854 incident MSKIs for the brigade, suggesting H2F providers evaluated 44% (2,140/4,854) of all incident MSKIs during this time. The most common clinic codes for these MSKIs were from physical therapy (43%; $n=2,091$), primary care (25%; $n=1,197$), emergency room (6%; $n=297$), and diagnostic radiology (3%; $n=166$).

Table 1. Population Characteristics by Person-Years, U.S. Army H2F-Resourced Stryker Brigade Combat Team, October 2021-September 2022

	Brigade total (4,780 p-y)	Battalion type				
		Cavalry (578 p-y)	Engineer (571 p-y)	Field artillery (631 p-y)	Infantry ^a (2390 p-y)	Support ^b (610 p-y)
	%	%	%	%	%	%
Sex						
Female	10.9	8.9	18.5	12.5	6.9	19.5
Male	89.1	91.1	81.5	87.5	93.1	80.5
Age						
<25 years old	48.4	48.8	44.2	48.5	53.4	32.8
25-34 years old	41.4	42.1	44.7	41.6	39.3	45.3
≥35 years old	10.2	9.2	11.0	9.8	7.3	21.9
Rank						
Junior enlisted	56.9	53.9	54.5	52.4	63.2	42.6
NCO	27.7	30.6	28.6	29.1	26.2	28.4
Senior NCO, all officers	15.4	15.5	16.9	18.5	10.6	29.0
Race						
American Indian/Alaska Native	1.1	0.5	1.2	1.1	1.3	1.0
Asian	7.4	7.4	7.2	6.8	7.2	9.1
Black or African American	16.2	17.1	20.1	14.9	13.6	23.4
Native Hawaiian/Pacific Islander	2.5	3.4	1.8	1.9	2.2	3.9
White	70.6	70.6	67.0	72.4	74.4	57.8
Unknown/Not applicable	2.1	1.1	2.6	2.8	1.3	4.9
Ethnicity						
Hispanic or Latino	19.1	19.0	21.8	18.0	18.8	19.0
Not Hispanic or Latino	80.9	81.0	78.2	82.0	81.2	81.0
Total		12.1	12.0	13.2	50.0	12.8

Abbreviations: H2F, Holistic Health and Fitness; NCO, noncommissioned officer; p-y, person-years.

^aThree infantry battalions are represented in this column.

^bWe included the brigade headquarters element with the brigade support battalion (BSB). We included the BSB forward support companies with the battalions to which they were functionally attached.

Characteristics of MSKI

The median time from MSKI onset to evaluation by an H2F healthcare provider was 5 days (IQR: 2-14; [Table 2](#)). When comparing evaluating provider type, the median time from MSKI onset to evaluation was 5 days for both PTs (IQR: 2-13) and ATs (IQR: 1-14). Over 40% of all MSKIs were evaluated within 3 days, and nearly two-thirds were evaluated within 7 days.

Upon presentation, 86% of the 2,140 MSKIs were self-reported to have caused individuals to miss or modify work or training. After evaluation, less than half of all MSKIs (41%) were assigned any limited duty restriction (time-loss). When limited duty was assigned, the median duration was 7 days (IQR: 3-7). The youngest and most junior-ranking personnel were more likely to have an MSKI resulting in time-loss ([Table S1](#)). For battalion, sex, race, and ethnicity, the percentage of time-loss and non-time-loss MSKI was comparable.

Half of all MSKIs were reported to have occurred during unit physical (fitness) training, the most common activity at time of injury ([Table 2](#)). Two-thirds of all MSKIs were to the spine (including neck), knee, and foot and ankle, and 43%, 54%, and 52% of these MSKIs, respectively, occurred during unit physical training. Only elbow, wrist, and hand injuries occurred most during personal time (30%). Lastly, ATs evaluated most MSKIs for each body region, consistent with staffing numbers.

DISCUSSION

This is the first study to report incidence rates for MSKIs evaluated by H2F healthcare providers in an Army brigade. We further characterized aspects of MSKI not typically available in Army systems²⁴ using injury tracker data collected by the H2F team. We highlight 4 key sets of findings and describe how they contribute to the literature and lay the groundwork for subsequent comparative research assessing the effect of H2F on MSKI and time-loss outcomes.

First, we were able to estimate MSKI rates by battalion type within this brigade combat team and standardize for differences in the distributions of age, sex, and rank by battalion. Standardization did not substantively alter battalion estimates, suggesting other factors are responsible for the differences in MSKI rates between battalions. These could include differences in physical fitness levels,²⁵ physical training volume and intensity,²⁶ and military occupations,^{16,27} each of which has been associated with MSKI in prior research. Our findings did contrast research that has reported female sex and older age have greater risk for MSKI.^{2,28,29} This could possibly be influenced by occupation, which these data did not contain, or the smaller relative study numbers for female sex and older age categories. However, other research found that female soldiers are not at greater risk of injury relative to male soldiers when controlling for covariates including occupational physical demands, body composition, or physical performance.³⁰ Comparable with a prior study during an

Table 2. Characteristics and Outcomes of Musculoskeletal Injuries (MSKIs) Evaluated by H2F Healthcare Providers, U.S. Army H2F-Resourced Stryker Brigade Combat Team, October 2021-September 2022

	All MSKIs ^a (n = 2,140)	Musculoskeletal injury (MSKI) by body region						
		Spine including neck (n = 536)	Knee (n = 466)	Foot and ankle (n = 446)	Shoulder and upper arm (n = 229)	Hip and upper leg (n = 204)	Lower leg (n = 181)	Elbow, wrist, and hand (n = 78)
	%	%	%	%	%	%	%	%
Time to evaluation								
Same day	4.5	6.0	2.4	6.7	1.3	8.8	1.1	1.3
1-3 days	37.9	40.9	36.3	41.9	31.0	35.8	33.1	41.0
4-7	21.0	20.3	22.5	18.6	22.3	25.0	19.3	20.5
8-14	14.3	11.4	14.8	13.9	18.3	12.7	18.2	16.7
15-30	9.2	6.9	9.4	7.0	15.7	8.3	12.2	11.5
31-59	5.0	6.0	5.4	4.5	4.4	1.0	7.7	3.8
>60	8.1	8.6	9.2	7.4	7.0	8.3	8.3	5.1
Median days [IQR]	5 [2-14]	4 [1-12]	5 [2-14]	4 [1-11]	7 [2-16]	4 [1-9]	7 [2-17]	4 [1-14]
Initial time-loss status								
No time-loss	58.8	64.0	64.2	50.7	65.9	52.9	49.2	50.0
Any time-loss	41.2	35.6	35.8	49.3	34.1	46.6	50.3	50.0
Median days [IQR] ^b	7 [3-7]	5 [3-7]	7 [4-7]	7 [3-7]	7 [3-7]	6 [3-7]	7 [4-7]	7 [3-7]
Injury activity								
Unit physical training ^c	50.5	42.7	54.1	51.6	38.4	64.7	71.8	25.6
Personal time	16.1	19.0	13.9	16.4	21.8	9.8	6.1	29.5
Unit military training	15.4	12.7	18.2	18.2	17.5	9.8	11.0	19.2
Non-specific onset	12.4	17.2	10.1	7.4	15.7	12.7	11.0	14.1
Work-related nontraining	5.4	8.4	3.2	6.3	6.1	2.5	0	11.5
Battalion								
Cavalry	10.6	13.4	9.2	10.5	12.7	6.4	9.9	5.1
Engineer	12.1	10.4	13.1	11.7	10.9	15.2	15.5	9.0
Field artillery	9.7	8.8	11.8	9.4	9.6	10.3	11.0	1.3
Infantry	56.9	55.8	55.6	57.2	58.5	57.4	50.8	79.5
Support	10.7	11.6	10.3	11.2	8.3	10.8	12.7	5.1
H2F evaluating provider								
Physical therapist	36.8	46.1	33.0	33.9	40.2	40.2	28.7	13.0
Athletic trainer	63.0	53.4	67.0	66.1	59.8	59.8	71.3	87.0
Total^a		25.0	21.8	20.8	10.7	9.5	8.5	3.6

Abbreviations: H2F, Holistic Health and Fitness; IQR, interquartile range; MSKI, musculoskeletal injury.

^a2,140 MSKIs sustained by 1,722 soldiers. Five MSKIs missing injury activity, 4 missing time-loss status, and 3 missing H2F evaluating provider type were not included in the percentages for those variables.

^bThis is the median duration of time-loss when time-loss was assigned. Most MSKIs did not require time-loss (median=0, IQR [0-5]).

^cOrganized physical fitness training, typically occurring each morning at unit echelons.

operational exercise,³¹ we found senior rank had lower MSKI rates, suggesting a potentially protective nature of rank in a hierarchical military structure. For race, although we did see large differences for some minoritized race and ethnic groups, these could be related to other factors that we could not analyze. For example, a prior study in active Army soldiers showed odds ratios for low back injury by race became much closer to the referent (White) after adjusting for age, sex, rank, occupation, body mass index, and tobacco use.³² Future MSKI studies in H2F-resourced brigades should measure additional factors like these.

Second, the self-reported date of onset was recorded for each MSKI, allowing us to estimate access to care from the date of injury onset to evaluation. In the Military Health System (MHS), access to care is measured from appointment request date to appointment, with the standard for specialty care (e.g., physical therapy) being 4 weeks from request.³³ Our reported 5-day median from MSKI onset to evaluation parallels 2 small studies' findings comparing direct access PT with PT referral

via primary care manager (PCM) in active-duty personnel with ankle injuries.^{9,34} A prior review suggests early access to physical therapy treatment in military populations might accelerate improvement compared with traditional care.³⁵ Our results suggest embedding H2F healthcare providers may reduce the number of days from MSKI onset to evaluation in H2F-resourced units.

Third, injury activity was recorded for all MSKIs, irrespective of time-loss status. Although injury activity is documented in systems capturing military duty limitations (e.g., e-Profile),²⁴ this is not typically available for non-time-loss MSKI. A recent publication reported injury activity or external cause codes are documented in only 10% of incident injuries across MHS.³⁶ We found half of MSKIs were attributed to unit physical training, which has been previously observed to be associated with MSKI.^{36,37} Future research should assess the impact of H2F strength and conditioning coaches (SCCs) designing physical training programs on mitigating MSKI, especially those considered overuse.

Musculoskeletal injury (MSKI) rates per 1000 person-months (95% CI)

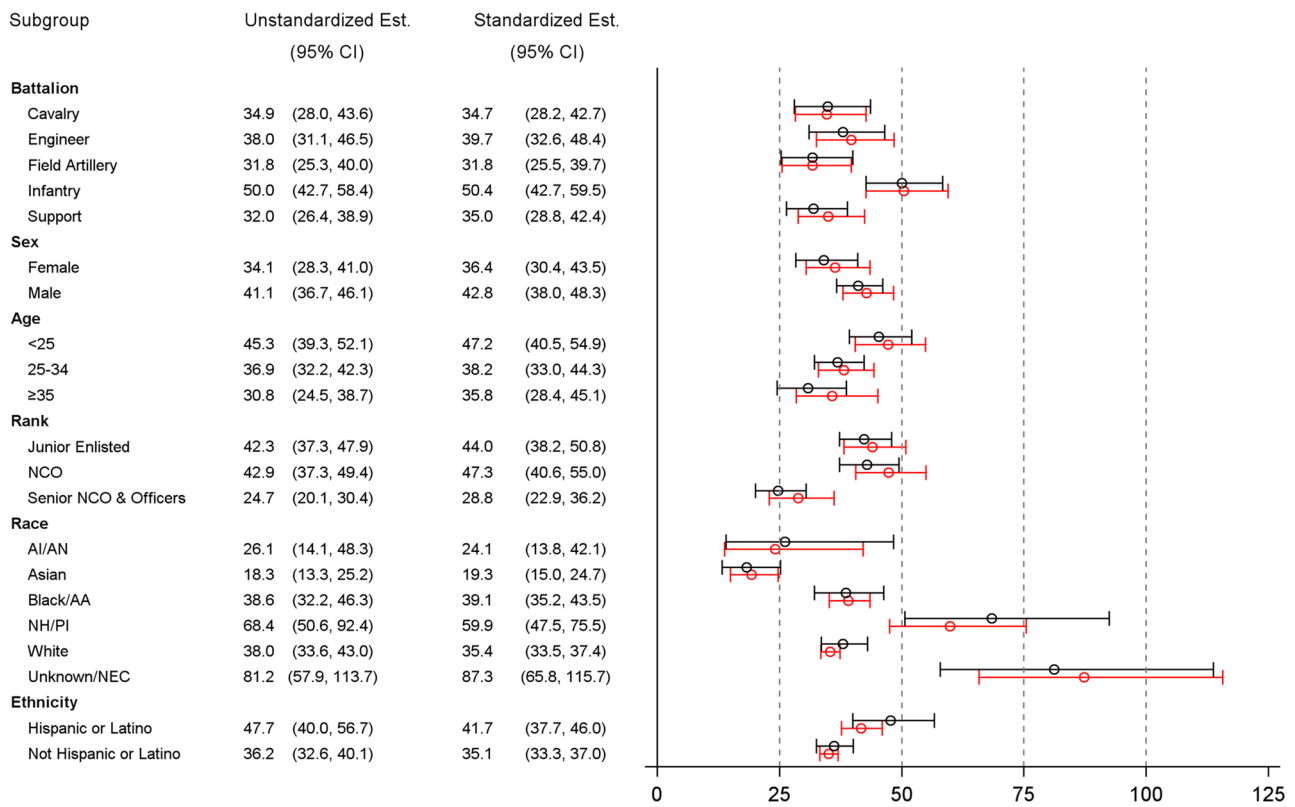


Figure 1. Forest plot of unstandardized (black; upper estimate for each covariate) and standardized (red; lower estimate for each covariate) musculoskeletal injury (MSKI) rate estimates stratified by covariates. Standardized rates used the marginal population distributions of battalion, sex, age and rank of the brigade overall. Abbreviations: AA, African American; AI/AN, American Indian/Alaska Native; Est., estimate; NCO, noncommissioned officer; NEC, not elsewhere classified; NH/PI, Native Hawaiian/Pacific Islander.

Fourth, the assignment of time-loss, or absence of, was recorded for all MSKI evaluations. Matching limited duty restrictions with specific medical encounters is normally challenging because different systems house these data.²⁴ We were able to compare time-loss assignment with functional limitations reported at initial evaluation (Figure 2). Notably, although most MSKIs led individuals to miss or modify work or training before evaluation, H2F healthcare providers managed most of these MSKIs without imposing formal duty limitations. Prior research conducted in demographically similar infantry brigade combat teams found soldiers reported over 60% of all injuries, most of which were MSKIs, required limited duty days.³⁷ We found only 41% of MSKIs were assigned time loss at initial evaluation. Although we could not measure cumulative time-loss, prior research reported the mean total time-loss duration of soldiers with ankle injuries managed by H2F PTs and ATs was approximately 25% of the duration when compared with PCMs.⁹

STUDY LIMITATIONS

As with all research, there were some limitations to our study. First, because of data deidentification, we were unable to link

individual encounters to the EHR or e-Profile. This limited our ability to validate MSKI, assess prior injury, and measure cumulative time-loss. Second, H2F MSKI rates reported in this study appear to underestimate the overall rates of the brigade based on the available EHR data. Reasons contributing to this could include external healthcare providers (e.g., primary care and emergency) documenting MSKIs that were never evaluated by the H2F team, ongoing MSKI specialty care (e.g., orthopedics) appearing as incident in the EHR, and H2F OT MSKI evaluations not being incorporated in the injury tracker during the study period. Lastly, additional variables related to MSKI, such as physical training volume and intensity, or occupation were not available. These could help better understand MSKI in H2F-resourced brigades and shape subsequent research and prevention efforts.

SIGNIFICANCE OF ONGOING H2F RESEARCH

Since 2020, brigade combat teams and other brigades (e.g., combat aviation, medical, sustainment) began receiving up to 35 H2F personnel to improve the health and readiness of 2,000-5,000 active-duty personnel in these units (Figure S1).⁷ In December 2024, the number of personnel in H2F-resourced brigades was

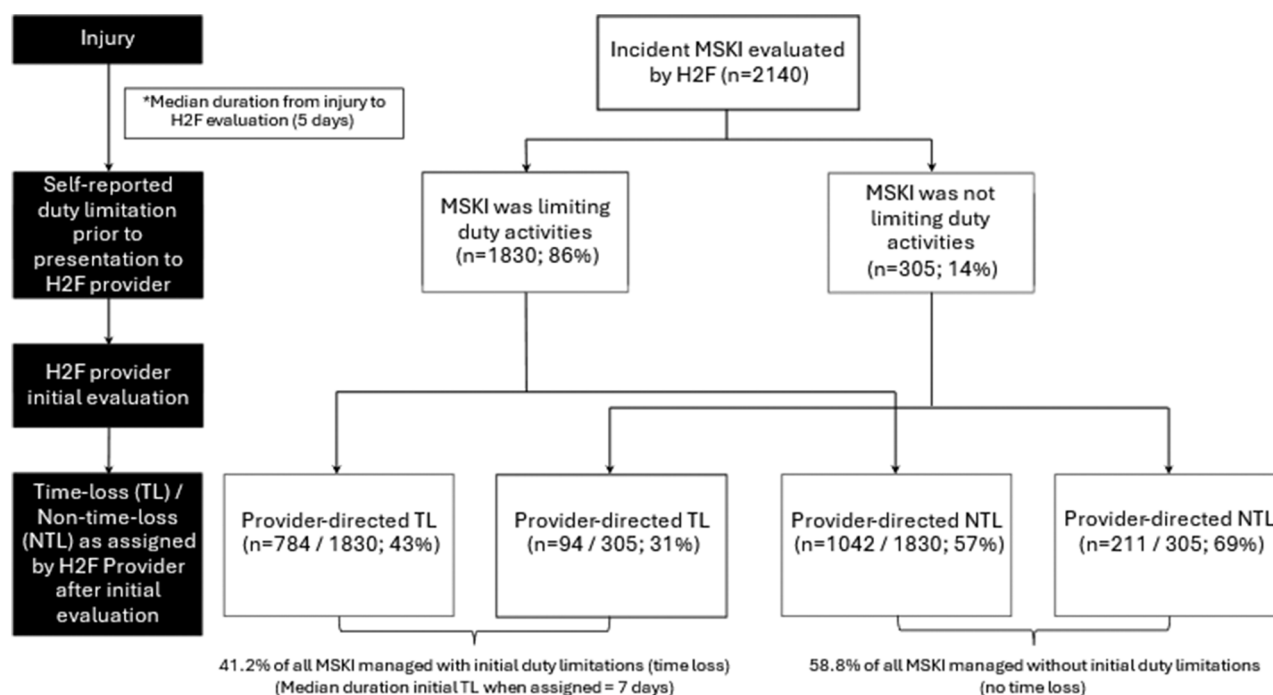


Figure 2. Self-reported functional limitations at initial evaluation versus Holistic Health and Fitness (H2F) provider-assigned limited duty. Abbreviations: MSKI, musculoskeletal injury; NTL, non-time-loss; TL, time-loss.

reduced from 35 to 22 personnel, decreasing ATs from 7 to 4 and SCCs from 14 to 7 personnel. At the time of writing, it was announced that future H2F teams (October 2025 onward) would not include ATs, receiving 4 additional SCCs instead. In this study, ATs evaluated nearly two-thirds of all MSKIs evaluated by H2F. Changes in AT staffing may affect the outcomes associated with the H2F system and should be evaluated closely.

CONCLUSION

This is the first study to report MSKI rates in an H2F-resourced brigade combat team in the U.S. Army. Using data collected by H2F personnel not routinely found in Army systems, we were able to characterize MSKIs evaluated by H2F PTs and ATs. Our findings are consistent with the hypothesis that the H2F system has expedited access to MSKI evaluation and management, most of which occurs without time-loss. As the H2F system evolves, we recommend caution in generalizing these findings to brigades and H2F teams of differing composition. This descriptive observational study will inform the findings from future research aiming to evaluate the effect of H2F implementation on MSKI outcomes.

ACKNOWLEDGMENTS

None declared.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Military Medicine* online.

FUNDING

Dr Marshall's contributions to this research were partly supported by a CE19 Injury Control Research Center award (R49CE003092) from the National Center for Injury Prevention and Control at the Centers for Disease Control and Prevention (CDC).

CONFLICT OF INTEREST STATEMENT

None declared.

DATA AVAILABILITY

The authors do not own the data used in this study. Requests for these data must be made through the Defense Health Agency (DHA) and the U.S. Army.

INSTITUTIONAL REVIEW BOARD

This research was determined Exempt by the Medical Center of Excellence, United States Army Institutional Review Board (23-00010e), and the University of North Carolina at Chapel Hill Institutional Review Board (23-1900).

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE

Not applicable.

INSTITUTIONAL CLEARANCE

This manuscript underwent review by the Defense Centers for Public Health—Aberdeen and was approved to submit for external publication.

INDIVIDUAL AUTHOR CONTRIBUTION STATEMENT

J.A.N. conceptualized the study and methodology, analyzed the data, and authored the original draft with assistance from M.J.F. and S.W.M. S.I.R., M.C.C., and E.M.W. assisted with the methodology and writing. C.V.O. designed the injury tracker and procured the data with M.K.B. All authors provided feedback on the manuscript and approved the final version. S.W.M. provided oversight of the entire process.

DECLARATIONS

J.A.N., M.K.B. and C.V.O. are active-duty members of the United States Army. M.C.C. is employed by the Defense Health Agency.

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